

AI Club Project Report
On

Smart Retail Stock Management

Submitted to **CMREC, HYDERABAD**

In Partial Fulfilment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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CMR ENGINEERING COLLEGE

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(Accredited by NAAC, Approved by AICTE NEW DELHI, Affiliated to JNTU, Hyderabad)

(Kandlakoya, Medchal Road, R.R. Dist. Hyderabad-501 401)

(2025-2026)

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CERTIFICATE

This is to certify that the project entitled “**Face Recognition System for Attendance**” is a bonafide work carried out by

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in partial fulfilment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **Computer Science and Engineering** CMR Engineering College, affiliated to JNTU, Hyderabad, under our guidance and supervision.

The results presented in real time project have been verified and are found to be satisfactory. The results embodied in this project have not been submitted to any other university for the award of any other degree or diploma.

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DECLARATION

This is to certify that the work reported in the present project entitled **“Smart reatail Stock Management”** is a record of bonafide work done by us in the Department of Computer Science Engineering, CMR Engineering College, JNTU Hyderabad. The reports are based on the project work done entirely by us and not copied from any other source. We submit our project for further development by any interested students who share similar interests to improve the project in the future.

The results embodied in major project report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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ACKNOWLEDGEMENT

We are extremely grateful to **Dr. A Srinivasula Reddy**, Principal and **Dr. Sheo Kumar**, HOD, **Department of CSE, CMR Engineering College** for their constant support. We are extremely thankful to **Mr. Abdul Mujeeb** Assistant Professor, Internal Guide, and **Dr. Mrutyunjaya S Yalawar**, Associate professor, AI Club Coordinator, Department of CSE, for his constant guidance, encouragement and moral support throughout the project. We will be failing in duty if we do not acknowledge with grateful thanks to the authors of the references and other literatures referred in this Project. We express our thanks to all staff members and friends for all the help and co-ordination extended in bringing out this project successfully in time. Finally, we are very much thankful to our parents who guided us for every step.

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ABSTRACT

Smart retail stock management leverages advanced technologies such as IoT, AI, and automation to optimize inventory management. Traditional retail systems often struggle with inefficiencies like stockouts, overstocking, and manual inventory tracking. By adopting smart systems, retailers can enhance stock visibility, reduce operational costs, and improve the overall shopping experience. This system integrates real-time data collection, predictive analytics, and automated replenishment to ensure accurate stock levels, reduce human error, and streamline the supply chain. The retail industry has undergone a rapid transformation in recent years with the integration of IoT (Internet of Things) technology. IoT technology allows retailers to gain insights into customer behavior, optimize inventory management, and improve the overall shopping experience. This paper presents the development of a smart retail management system using IoT technology. The proposed system is designed to improve the efficiency and effectiveness of retail operations. The system consists of various IoT devices such as RFID readers, sensors, and cameras that collect data on customer behavior, inventory levels, and store conditions. The data collected is analyzed using machine learning algorithms to provide retailers with valuable insights into customer behavior, demand trends, and inventory levels. These insights can be used to optimize store operations, improve inventory management, and provide a better shopping experience for customers. The system also includes a mobile application that enables customers to access information about products, promotions, and discounts. The application uses beacon technology to provide personalized offers to customers based on their location and purchase history. Additionally, the system includes a dashboard for store managers that provides real-time data on inventory levels, sales trends, and customer traffic. The proposed system has the potential to revolutionize the retail industry by providing retailers with valuable insights into customer behavior and optimizing store operations. The system is scalable and can be easily adapted to meet the needs of different types of retail stores. Overall, this paper demonstrates the potential of IoT technology in the retail industry and provides a roadmap for the development of smart retail management systems.

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Chapter 1: INTRODUCTION

Background of Study

Retail businesses face significant challenges in managing inventory efficiently due to fluctuating customer demand, supply chain complexities, and human errors in stock tracking. Traditional inventory management systems rely on manual processes or outdated software, often leading to stock shortages, overstocking, and financial losses. In today's competitive market, retailers must adopt innovative solutions to ensure optimal stock levels, reduce operational costs, and enhance customer satisfaction. The emergence of smart technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and real-time data analytics has transformed stock management. Smart retail stock management systems integrate these technologies to provide real-time inventory tracking, demand forecasting, and automated restocking solutions. By leveraging these advancements, retailers can minimize losses, improve efficiency, and create a seamless shopping experience. This study explores the implementation of smart stock management in retail, analyzing its impact on efficiency, cost reduction, and customer service improvement.

Traditional retail stock management often faces challenges such as overstocking, stockouts, and inaccurate inventory tracking, leading to financial losses and customer dissatisfaction. With the advancement of technologies like IoT, AI, and real-time data analytics, smart retail stock management offers automated tracking, demand forecasting, and efficient restocking. This study explores how smart inventory systems can enhance efficiency, reduce costs, and improve overall retail operations.

Problem Statement

Retail businesses often struggle with inefficient stock management, leading to problems such as stock shortages, overstocking, inaccurate inventory tracking, and high operational costs. Traditional methods rely on manual tracking, which is prone to human errors, delays, and poor demand forecasting. These inefficiencies result in lost sales, increased storage costs, and dissatisfied customers due to unavailable products. Additionally, issues like theft, shrinkage, and supply chain disruptions further complicate inventory control. To address these challenges, a Smart Retail Stock Management system is needed— leveraging IoT, AI, and real-time analytics to optimize inventory tracking, automate restocking, and provide predictive insights. By integrating smart technologies such as barcode or RFID-based tracking, AI-powered demand forecasting, and real-time alerts for stock levels, retailers can streamline operations, reduce losses, and enhance customer satisfaction. This solution ensures products are available when needed, minimizes wastage, and improves overall business efficiency, making retail stock management smarter and more effective. In the rapidly evolving retail industry, efficient stock management is crucial for minimizing losses, reducing waste, and ensuring customer satisfaction. Traditional inventory tracking methods often lead to challenges such as overstocking, stockouts, inaccurate demand forecasting, and high operational costs. Smart retail stock management leverages advanced technologies like IoT, AI, and real-time data analytics to optimize inventory control, automate restocking processes, and improve supply chain efficiency. By implementing a smart system, retailers can gain real-time visibility into stock levels, predict demand accurately, and streamline operations, ultimately enhancing profitability and customer experience.



Main Objective

The primary objective of this project is to design and implement a smart retail stock management system that leverages advanced technologies to optimize inventory control, improve efficiency, and enhance decision-making. The system aims to address common stock management challenges such as overstocking, stock shortages, human errors, and inefficient supply chain operations.

The main objective of this project is to develop a smart retail stock management system that enhances inventory accuracy and optimizes stock levels using technologies like IoT, AI, and real-time data analytics. The system will automate stock tracking, predict demand, and streamline restocking to reduce overstocking and stock shortages. By improving efficiency and reducing operational costs, this project aims to enhance overall retail management and ensure better product availability for customers.

By achieving these objectives, the project will contribute to higher profitability, better resource utilization, and improved overall retail efficiency.

specific Objectives

Specific Objectives: Automate

Inventory Tracking – Utilize IoT sensors, RFID tags, or barcode systems to monitor stock levels in real time and reduce human dependency.

Improve Demand Forecasting – Implement AI and machine learning algorithms to analyze past sales data, customer trends, and seasonal patterns for accurate stock predictions.

Enhance Stock Replenishment – Automate restocking alerts and supplier integration to maintain optimal inventory levels, preventing overstocking and stockouts.

Reduce Operational Costs – Minimize waste, storage expenses, and labor costs through data-driven stock management strategies.

Improve Customer Experience – Ensure product availability, reduce wait times, and enhance shopping convenience through better stock visibility.

Optimize Supply Chain Management – Integrate real-time data analytics to improve coordination between suppliers, warehouses, and retail outlets.

Purpose and Scope of Study

Purpose

The purpose of this project is to create a smart retail stock management system that improves inventory control, reduces losses, and enhances efficiency in retail operations. By leveraging IoT, AI, and real-time data analytics, the system will automate stock tracking, predict demand accurately, and optimize restocking processes. This will help retailers minimize overstocking and stock shortages, reduce operational costs, and improve customer satisfaction by ensuring product availability. Ultimately, the project aims to modernize retail stock management, making it more efficient, data-driven, and cost-effective.

The main scope deliverables of the project would be to:

- Automated Inventory Management– Utilize IoT, RFID, and barcode systems to track stock levels in real time.
- Demand Forecasting– Implement AI-based analytics to predict sales trends and optimize stock levels.
- Automated Stock Replenishment – Generate alerts and integrate with suppliers for efficient restocking.
- Data Analytics and Reporting – Provide insights on inventory performance, sales trends, and stock
- issues for better – Ensure product availability and reduce stock-related service.

LITERATURE REVIEW AND THEORY

Literature Review Retail stockmanagement has evolved from manual tracking to technology-driven solutions that enhance efficiency and accuracy. Traditional inventory methods often lead to challenges such as stock discrepancies, overstocking, and shortages due to human errors and outdated tracking systems. Studies highlight the role of IoT in providing real-time stock visibility through RFID, barcode scanning, and sensors, reducing losses and automating inventory updates. AI and machine learning further improve demand forecasting by analyzing sales patterns and predicting future stock needs, minimizing waste and optimizing restocking. Additionally, data analytics helps retailers make informed decisions by tracking sales trends and customer behavior. Automated stock replenishment and supply chain integration ensure timely restocking, reducing lead times and improving operational efficiency. This literature review emphasizes the benefits of smart inventory management systems, demonstrating their potential to enhance retail operations, reduce costs, and improve customer satisfaction.

Effective stock management is essential for retail businesses to maintain optimal inventory levels and meet customer demands. Traditional methods, which rely on manual tracking and periodic audits, often result in inefficiencies, stock shortages, and financial losses. Research highlights the role of IoT in enabling real-time inventory tracking through RFID and barcode scanning, reducing errors and improving stock visibility. AI-powered demand forecasting helps retailers predict sales trends, optimize inventory, and prevent overstocking or stockouts. Additionally, big data analytics allows businesses to analyze purchasing patterns, streamline supply chains, and enhance decision-making. Automated restocking systems further improve efficiency by ensuring timely replenishment and reducing operational costs. This study explores the impact of these smart technologies on modern retail stock management, emphasizing their role in improving accuracy, efficiency, and overall pro

Online shopping sites like Amazon, Alibaba, and Walmart use AI, IoT, and data analytics for efficient stock management. Books like "Retail Management" by Berman & Evans and "Supply Chain Management" by Sunil Chopra provide insights into inventory optimization. These resources help understand smart stock management strategies for retail businesses.

Existing System



Figure 1: Traditional methods

Strength

The project ensures real-time inventory tracking using IoT and AI, reducing stock errors. It improves demand forecasting, preventing overstocking and shortages. Automated stock replenishment helps in timely restocking, enhancing efficiency. The system reduces operational costs by minimizing waste and manual labor. Ultimately, it enhances customer satisfaction by ensuring product availability.

Weakness

The project has a high initial cost, as implementing IoT, AI, and automation requires significant investment. It involves technical complexity, needing skilled personnel for setup and maintenance. Data security risks arise due to potential cyber threats in online inventory management. The system is dependent on internet and power, making it vulnerable to outages. Integration challenges may occur when connecting with existing retail systems. Hardware and software malfunctions could disrupt operations and require timely troubleshooting. Additionally, small retailers may find adoption difficult due to resource constraints.

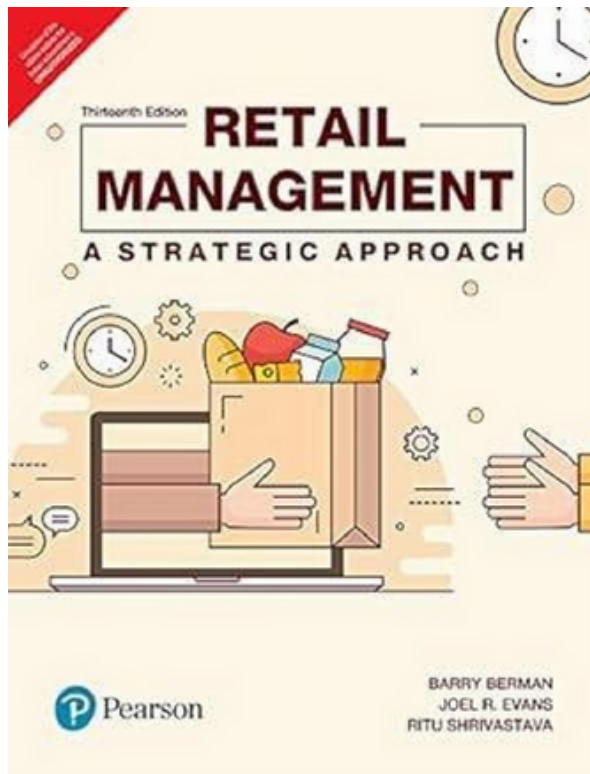


Figure 2: Books

"Retail Management: A Strategic Approach" – By Barry R. Berman & Joel R. Evans -Covers inventory control, retail operations, and stock management strategies. "Supply Chain Management: Strategy, Planning, and Operation" – By Sunil Chopra & Peter Meindl- Discusses inventory optimization, supply chain efficiency, and demand forecasting.

Strength

Theselected books provide comprehensive insights into retail stock management, covering inventory control, supply chain optimization, and demand forecasting. They include real- world case studies, helping readers understand practical applications in successful businesses. Many books explore modern technologies like AI, IoT, and data analytics, showing how they enhance stock management. They strike a balance between theory and practice, making them valuable for both academic research and real-world implementation. Written by industry experts and scholars, these books offer credible, well-researched knowledge essential for understanding smart retail stock management.

Weakness

Somebooks may contain outdated information, as retail technology evolves rapidly. They can be theory-focused, lacking hands-on implementation details for smart stock management. Additionally, books often provide generalized solutions, which may not fully address specific

CHAPTER :2 SCOPE OF THE PROJECT

2.1 General:

The project aims to introduce and democratize the latest machine learning technologies for retail investors. No prediction is 100% accurate. Therefore, the upper bound and lower bound of the stock prices will be displayed to illustrate the trading range the investors should be looking at. This application serves as a supplementary quantitative tool for investors to see the market at a different perspective with the help of technology. This project is divided into 2 parts, namely a search component and an application component, aiming to provide retail investors with stock price predictions.

2.2 EXISTING SYSTEM :

The existing system relies on traditional methods of stock market analysis, which often lack the accuracy needed for effective decision-making by investors. These methods may include basic technical analysis, fundamental analysis, and simple statistical models. However, they may fall short in accurately predicting stock market movements, leading to potential losses for investors. Inadequate Feature the finance world stock trading is one of the most important activities. Stock market prediction is an act of trying to determine the future value of a stock or other financial instrument traded on a financial exchange. This paper explains the prediction of a stock using Machine

Learning. The technical and fundamental or the time series analysis is used by the most of the stockbrokers while making the stock predictions. The programming language is used to predict the stock market using machine learning is Python. We propose a Machine Learning (ML) approach that will be trained from the available stocks data and gain intelligence and then uses the acquired knowledge for an accurate prediction. In this context, this study uses a machine learning technique called Support Vector Machine (SVM) to predict stock prices for the large and small capitalizations and in the three different markets, employing prices with both daily and up to the minute frequencies.

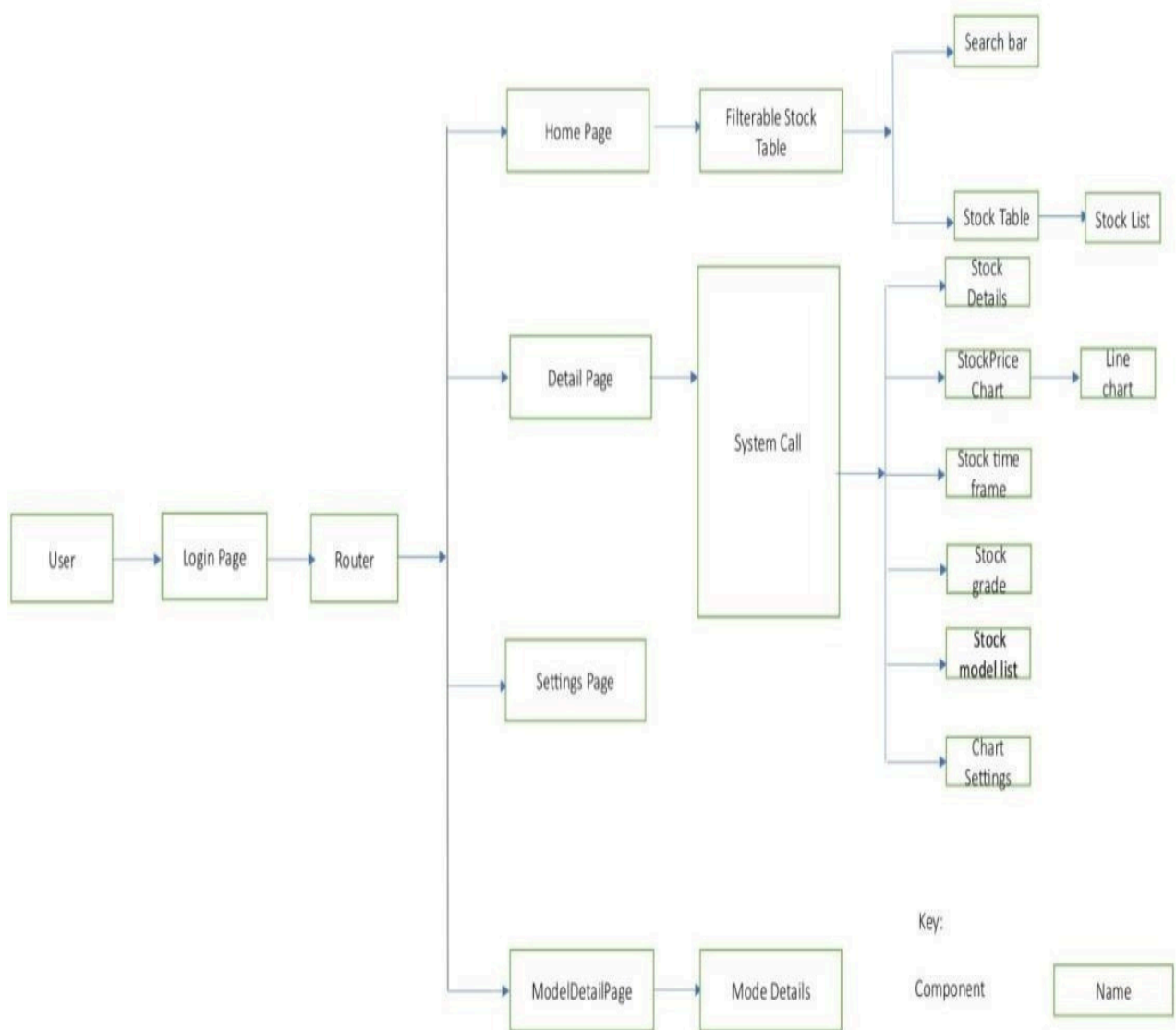
2.3 PROPOSED SYSTEM :

The proposed system introduces the use of the Support Vector Machine (SVM) algorithm for stock price prediction. SVM offers advantages such as better accuracy, capacity control of the decision function, and the ability to utilize kernel functions for complex data. By implementing SVM, the system aims to enhance the predictability of financial movements, providing more reliable recommendations for both short-term and long-term investors. This project, we have made a time-series analysis and it doesn't need n-fold cross validation. Methodology since it's sequential data. We split our dataset in train and test data. Top 80 percent of data will be Train data and the remaining will be test. The prediction model, which is based on SVM and independent analysis, combined called SVM-ICA, is proposed for stock market prediction. Various time series analysis models are based on machine learning. The SVM designed to solve regression problems in nonlinear classification and time series. In this proposed system, we focus on predicting the stock values using machine learning algorithms like Random Forest and Support Vector machines.

ADVANTAGES:

- Capacity Control -
- >Improved Accuracy -
- >Kernel Functions -
- >Comparative Analysis
- s ->Data Source -
- >Capacity Control -
- >Improved Accuracy -
- >Kernel Functions -
- >Comparative Analysis
- s ->Data Source

2.4 SYSTEM ARCHITECTURE :



CHAPTER 3 PROJECT DESCRIPTION

3.1 GENERAL

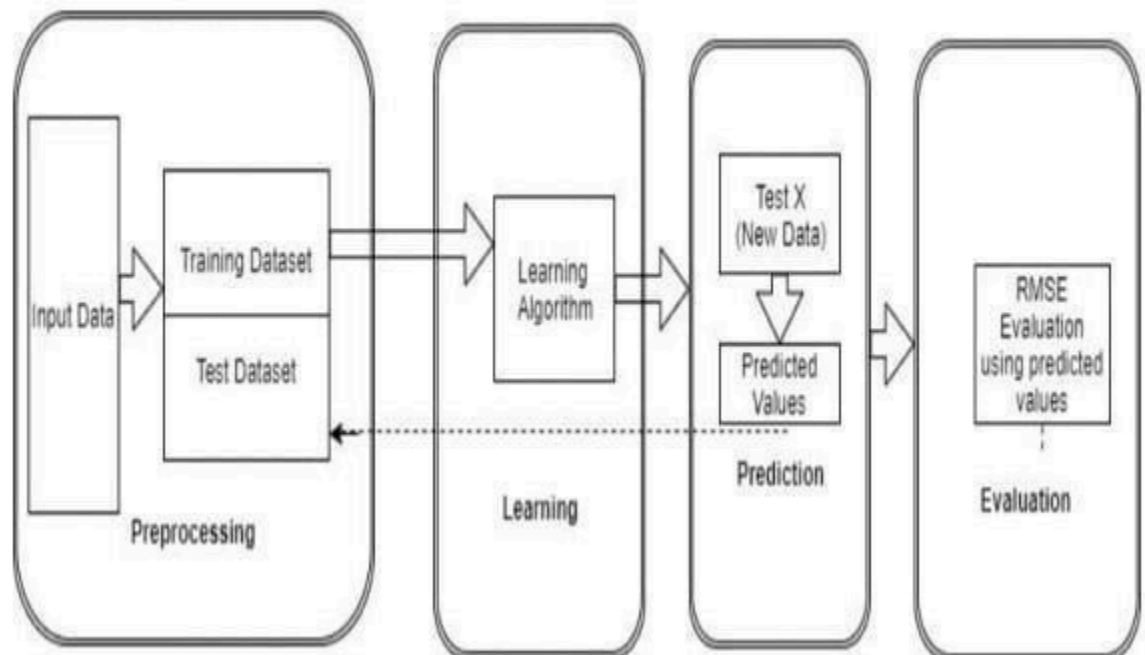
The develop a predictive model that can forecast stock prices using historical data and machine learning techniques. The primary goal is to provide accurate predictions that can assist investors and traders in making informed decisions.

3.2 MODULES NAME

This project having the following 5 modules:

- ★ Data collection
- ★ data preprocessing
- ★ model training
- ★ real time prediction
- ★ evaluation
- ★ deployment

3.3 Modules Explanation



● Modules-Connectivity Diagram

The module diagram is an expanded data flow diagram used to define a system's business logic. It is composed of the following key elements: Components: Components, such as computation or message displays, are functional components in the system. These elements have input and output ports, which allow data to be transmitted between them and are represented as rectangles.

Data Flow:

Flow lines indicate the movement of data within a component. These lines define the order in which elements are implemented, as well as the amount of data that flows through the system. Depending on the nature of data, they can be represented

Control Flow Constructs:

Control flow constructs, such as those used with Schneiderman boxes, allow you to design module logic in accordance with Structured Programming concepts, making them a more pleasant addition to your module diagram. These constructs include timed execution, looping, dependent branching, and .

Module Events:

The module diagram will cover one or more modules, and each of these events is displayed separately within it. By clicking on an event or selecting a group of events, the user can view the specifics of each module event. It is a module event, as you can tell by looking underneath the module name. Module Events can be triggered by a variety of events, such as the beginning of a related module, interacting with your user interface, selecting an override button, or using internal triggers within this module

- Data Collection:

Modules: Use libraries like pandas for data manipulation, numpy for numerical operations, and yfinance or Alpha Vantage for fetching historical stock price data.

- Data Preprocessing: Modules: Utilize pandas for data cleaning, handling missing values, and feature engineering.

- Feature Engineering: Modules: Use libraries like ta (Technical Analysis Library in Python) for calculating technical indicators.
- Model Selection: Modules: Use scikit-learn for traditional machine learning models or tensorflow/keras for deep learning models.
- Model Training: Modules: Utilize scikit-learn or tensorflow/keras for model training.
- Real-Time Prediction: Modules: Use libraries like yfinance for fetching real-time stock price data and schedule for scheduling tasks.
- Evaluation: Modules: Utilize evaluation metrics from scikit-learn or custom metrics for financial data.

- Deployment:

Modules: Use frameworks like Flask or FastAPI for building APIs, or deploy as a web application using Django or Streamlit.

3.4 TECHNIQUE USED OR ALGORITHM USED

MACHINE LEARNING MODELS USED: FB Prophet
Python for backend
Flask framework for integration of frontend and backend
JavaScript framework with CSS and HTML for front end

CHAPTER 4 REQUIREMENT

4.1 GENERAL

These are the requirements for doing the project. Without using these tools & software's we cannot do the project. Therefore, we have two requirements to do the project. They are ->Hardware Requirements. ->Software Requirements.

4.2 HARDWARE REQUIREMENTS:

->Processor: Intel Core i5 or higher ->RAM: 8GB or higher ->Storage: 500GB HDD or SSD ->Operating System: Windows 10 or Linux ->By incorporating the proposed system with SVM Algorithm, investors can ->potentially improve their Investment decisions and maximize profits while ->Minimizing financial risks.

4.3 SOFTWARE REQUIREMENTS:

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks, tracking the teams, and tracking the team's progress throughout the development activity. ->Programming Language: Python ->Data Analysis Libraries: Pandas, NumPy ->Machine Learning Libraries : Scikit-learn ->Data Visualization: Matplotlib, Seaborn ->Integrated Development Environment (IDE): google

CHAPTER 5 IMPLEMENTATION

5.1 GENERAL

The implementation is nothing but source code of project

5.2 IMPLEMENTATION CODING:

```
import streamlit as st
import yfinance as yf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error
# Sample portfolio (symbol, quantity)
portfolio = {
    "AAPL": 10,
    "MSFT": 5,
    "GOOG": 7,
    "TSLA": 8,
    "AMZN": 6
}
st.title("📊 Stock Portfolio Dashboard")
# Fetch current prices
@st.cache_data
def get_current_prices(tickers):
    data = yf.download(tickers, period='1d', group_by='ticker', auto_adjust=True)
    prices = {}
    for ticker in tickers:
        try:
            prices[ticker] = data[ticker]['Close'][-1]
        except Exception:
            prices[ticker] = None
    return prices
tickers = list(portfolio.keys())
prices = get_current_prices(tickers)
# Create table
stock_table = pd.DataFrame({
    "Ticker": tickers,
    "Quantity": [portfolio[t] for t in tickers],
    "Current Price": [prices[t] for t in tickers]
})
st.dataframe(stock_table.set_index("Ticker"))
# Stock selection
selected_stock = st.selectbox("📈 Select a stock to view predictions:", tickers)
```

```

RandomForestRegressor(n_estimators=100,
# Function to get and process data def get_stock_data(ticker, start='2020-01-01',
end=e'2n0d2,4-01-01'):stock = yf.download(ticker, start=start,
=auto_adjust=True) return stock def prepare_data(data): data['Return']
=data['Close'].pct_change() data['SMA_10']
=data['Close'].rolling(window=10).mean() data['SMA_50']
data['Close'].rolling(window=50).mean() data['Target'] = data['Close'].shift(-1) data =
data.dropna() features = ['Return', 'SMA_10', 'SMA_50'] X = data[features] y =
data['Target'] return X, y, data # Only run prediction if a stock is selected if
selected_stock: stock_data = get_stock_data(selected_stock) X, y, full_data =
prepare_data(stock_data) X_train, X_test, y_train, y_test = train_test_split(X, y,
=test_size=0.2, shuffle=False) scaler = StandardScaler() X_train_scaled
scaler.fit_transform(X_train) X_test_scaled = scaler.transform(X_test) model =
random_state=42)
model.fit(X_train_scaled, y_train) y_pred = model.predict(X_test_scaled) rmse =
np.sqrt(mean_squared_error(y_test, y_pred)) mae = mean_absolute_error(y_test,
y_pred)
st.subheader(f"📈 Price Prediction for {selected_stock}")
st.write(f"**RMSE:** {rmse:.2f}, **MAE:** {mae:.2f}")
# Plot
fig, ax = plt.subplots(figsize=(10, 5))
test_dates = full_data.index[-len(y_test):]
ax.plot(test_dates, y_test.values, label='Actual Prices', color='blue')
ax.plot(test_dates, y_pred, label='Predicted Prices', color='red', linestyle='dashed')
ax.set_title(f'{selected_stock} - Actual vs Predicted Prices')
ax.set_xlabel("Date")
ax.set_ylabel("Price")
ax.grid(True)
ax.legend()
st.pyplot(fig)

```

Research Methodology

The first step in starting this project is gathering all of the necessary requirements in order to develop a website that is easy to maintain and meets user needs. The emphasis is primarily on study into various online bookstores that are already operational. Accordingly, questionnaires and interviews with target end users have been shown to be the most effective methods for gathering exact data from them. The design paradigm starts with a prototype of the graphical user interface (GUI) layer and then outlines the website's back end (database and system architecture) system, which is only visible to the portal's administrators. The prototype's construction was adjusted in response to end-user feedback in order to build an online book catalogue that meets the project's requirements at the completion. Iteration occurs as the prototype is refined to meet the user's needs while also allowing the developer to gain a better understanding of what needs to be done. The website is then deployed on a web hosting to make it available via the internet after the GUI design prototype is finalized.

Within a software project, the Software Development Life Cycle (SDLC) is a process that is followed for a software project. It is a detailed strategy that describes how to build, maintain, replace, and change or improve particular software. The life cycle is a mechanism for enhancing software quality and the development process itself. During the software development process, numerous software development life cycle models have been established and designed. To assure success in the software development process, each process model follows a set of stages specific to its type. Agile Methodology is the model I'm using for this project.

AGILE METHODOLOGY

To deliver a smooth execution of the project, a methodology has been chosen to assist in managing the project phase by phase, beginning with the initiation and ending with the closure. A methodology is necessary for selecting the best methods, practices, approaches, and procedures from among the many available. As a result, Agile Methodology was chosen for the development of the Online Book Store System platform. Agile technique is the ideal strategy to complete this project because it divides it into phases and requires constant adjustments and input until the final product is completed.

Agile methodology is an iterative project management strategy that allows a project to adapt to changing working conditions and develop in a short amount of time. In this method, which is referred to as "sprint," a project term is divided into shorter and repeatable stages. Sprint lengths were decided during the project's early planning stages and will be reflected in the final product.

Process

Research, Analysis and Gathering Requirement

This phase includes conducting literature reviews and analyses of existing systems in order to determine the requirements and features for an online book system website. Prepare documentation after evaluating the research study. Surveys and feedback are used to gather requirements. Requirements are subject to change over time. The prerequisites for the Online Book System platform are listed below:

A product catalogue on the home page

After a successful login, the user will be directed to this page. It will provide all of the book categories and provide a search keyword option for finding the desired title. It also has some unique elements, such as recommended titles and weekly special books

AdvancedSearch The user can utilize advanced search to find a book based on the title, author, category, and price range. The total number of books that match the specific search criteria will be presented. The user can choose a book and add it to their shopping cart from here.

AdvancedSearch The user can utilize advanced search to find a book based on the title, author, category, and price range. The total number of books that match the specific search criteria will be presented. The user can choose a book and add it to their shopping cart from here.

Book Description

If a user wants more information about a book, he/she can click on the title and be taken to a Book Description page. It includes annotations on the book's content as well as a link to Amazon.com where you can read the book review.

Advanced Search The user can utilize advanced search to find a book based on the title, author, category, and price range. The total number of books that match the specific search criteria will be presented. The user can choose a book and add it to their shopping cart from here.

Book Description

If a user wants more information about a book, he/she can click on the title and be taken to a Book Description page. It includes annotations on the book's content as well as a link to Amazon.com where you can read the book review.

Voting by users

A user can rate a book based on his or her interests. They can give it a five-star rating if it's excellent, four if it's very good, three if it's good, two if it's regular, and one if it's deficient. The final rating of a book is determined by the sum of all user ratings.

Shopping Cart

The user can manage a shopping cart that contains all of the books that they had chosen. The user can change, delete, and update the contents of their shopping cart. A final shopping cart summary is shown, which includes all of the products selected by the user as well as the overall cost.

Managing User Account

To access all of the website's functions, each user should create an account. The user can log in and out using the login and logout pages, respectively. The database will save all of the user sessions. Administration Special functions for the Administrator will be granted, such as adding or deleting a book category, adding or deleting a member, managing member orders, adding or deleting a debit/credit card type, and so on.

Planning Phase

Project Tasks/Weeks	1	2	3	4	5	6	7	8	9	10	11	12
Idea & Title Proposal												
Introduction												
Aim & Objective												
Analysis & Survey												
Methodology												
Proposal Defense												
Design & Development												
Interim Report Submission												

Project Tasks/Weeks	1	2	3	4	5	6	7	8	9	10	11	12
Design Phase												
Prototype Development												
Prototype Testing & Maintenance												
Prototype Finalization & Documentation												
Dissertation Submission												
Viva Presentation												

Table 2: Gantt Chart for FYP 2

Project Design Phase

In order to design a web site, the relational database must be designed first. The data model and the process model are the two aspects of conceptual design. The data model specifies which data should be stored in the database, whereas the process model defines how the data should be handled. To put it another way, the data model is used to construct the relational tables of a relational database. The process model is used to design the queries that will access those tables and perform operations on them.

A decomposition diagram illustrates a system's top-down functional decomposition and highlights its structure. The goal of the Functional Decomposition is to break down a system step by step, starting with the system's main function and progressing through the intermediate levels to the level of basic functions. More extensive process diagrams, such as Data Flow Diagrams (DFD) start with this diagram. Data Flow Diagrams illustrate the flow of data from external entities into the system, as well as from one process to the next. A DFD can be drawn using four symbols:

Rectangles that represent external entities such as data sources and destinations.

Ellipses, which represent processes that receive data as input, validate, process, and output it.

Data flows are represented by arrows, which can be electronic data or physical items.

Data stores, especially electronic stores such as databases, are represented by open- ended rectangles or a Disk symbol.

Development, Testing and Review Phase

Development Phase includes writing codes using programming languages. The website will be programmed using HTML and CSS. In the meantime, MySQL will be used as the database. This process will be repeated until the client is satisfied with the final product.

Maintaining Phase test the website's functionality. This process will take place in parallel with the development phase. This is to check that the programmed codes are working properly and that there are no errors in the coding.

Review Phase includes obtaining feedback from users and testers. This process will be followed once the finalized website has been created. This process is necessary in order to develop the platform in the future. For the evaluation process, this website will be shared with a group of users or testers.

CHAPTER 6 RESULTS & DISCUSSION

6.1 GENERAL The entire project is done on google colab using python in which html scripting language is used to run the application on a web page.

6.2 RESULTS Starting Page of Frontend

Here after visiting our Web Page, By clicking on the Start we will visit next page



CHAPTER 7 CONCLUSION

7.1 CONCLUSION

In conclusion, Python emerges as a powerful ally in the pursuit of accurate stock price prediction. By harnessing its capabilities for data analysis, machine learning, and visualization, researchers and analysts can develop sophisticated models that capture underlying patterns and trends in stock price movements. While challenges and uncertainties persist in financial markets, Python's flexibility and extensive ecosystem enable continuous refinement and validation of predictive models.

Ultimately, Python empowers stakeholders to make informed decisions, manage risks, and optimize their investment strategies in an event.

7.2 FUTURE ENHANCEMENT

Stock price prediction using the Support Vector Machine (SVM) algorithm in Python is a sophisticated approach that leverages machine learning techniques to forecast future stock prices. SVM is a powerful classification and regression tool that works by finding the hyperplane that best divides a dataset into classes. In the context of stock price prediction, SVM can be used for regression to predict continuous values, such as stock prices. In Python, this process typically involves using libraries such as scikit-learn for implementing the SVM algorithm, pandas for data manipulation, and matplotlib for visualization. The SVM model is then trained on the training set, and its performance is evaluated on the test set using metrics such as mean squared error (MSE) or R-squared. Fine-tuning the model's hyperparameters, such as the regularization parameter C and the kernel type (linear, polynomial, or RBF), can significantly enhance the prediction accuracy. Additionally, feature selection and engineering can further improve the model's performance by incorporating technical indicators like moving averages, RSI, and MACD.

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Gardiner.
4. The project focuses on algorithms such as ARIMA, Prophet and LSTM for
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7. Lipton. This project combines traditional technical analysis methods with
machine
8. learning algorithms to predict stock prices. It covers feature engineering and
model
9. selection using Python.
10. Figure 1: Traditional methods
11. Figure 2: Books

